Huddle Test Results

Date: 2026-09-29

Errors and Warnings

10031 TEMPERATURE ERROR: temperature 24.7 degrees below minimum threshold (-20 C).
10031 TEMPERATURE ERROR: temperature 69.6 degrees above threshold (60 C).
10031 MA POLARITY ERROR: most mA readings are not strictly positive; check current-sensor wiring polarity.
10032 MA POLARITY ERROR: most mA readings are not strictly positive; check current-sensor wiring polarity.
10033 MA POLARITY ERROR: most mA readings are not strictly positive; check current-sensor wiring polarity.
10035 MA POLARITY ERROR: most mA readings are not strictly positive; check current-sensor wiring polarity.

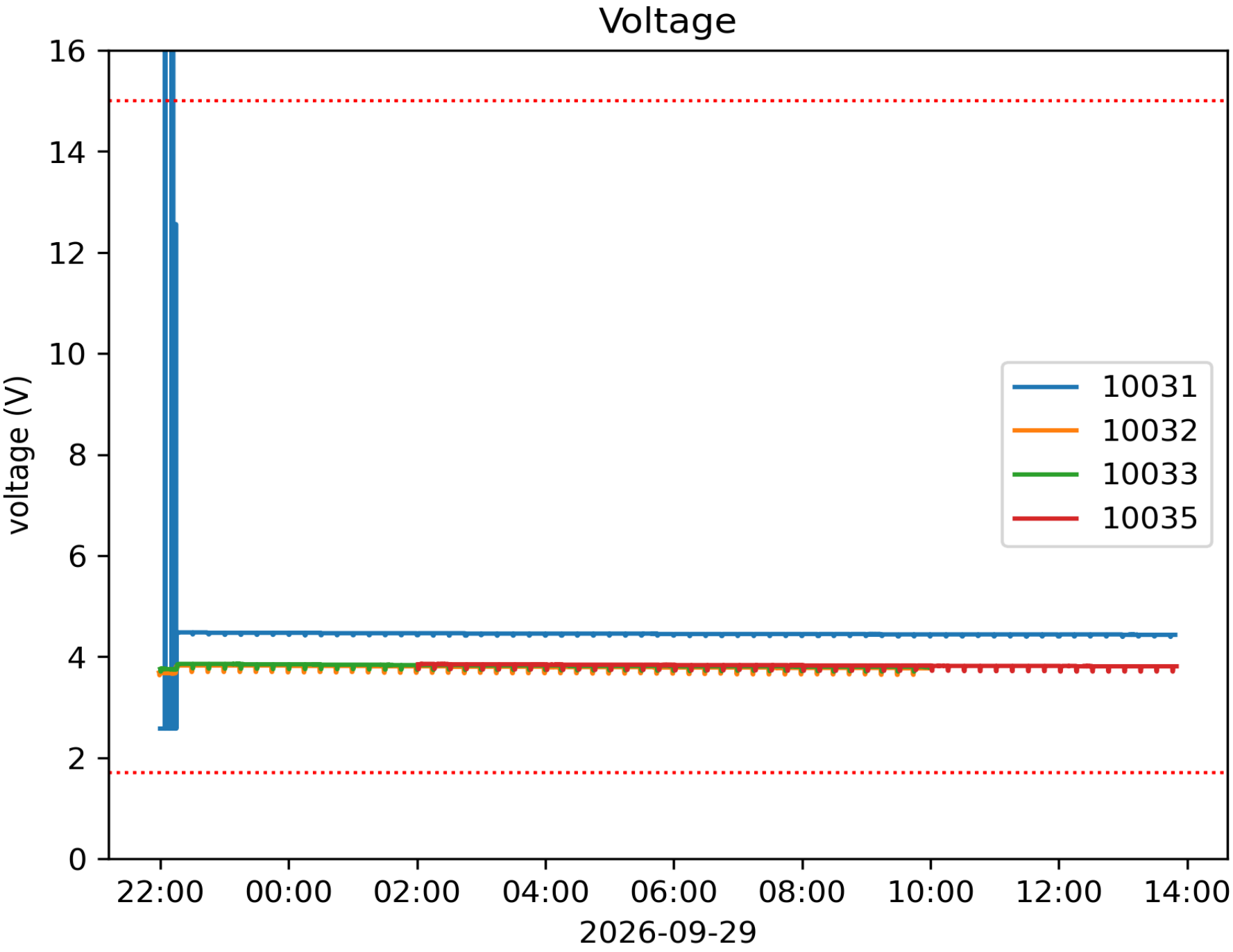
Notes

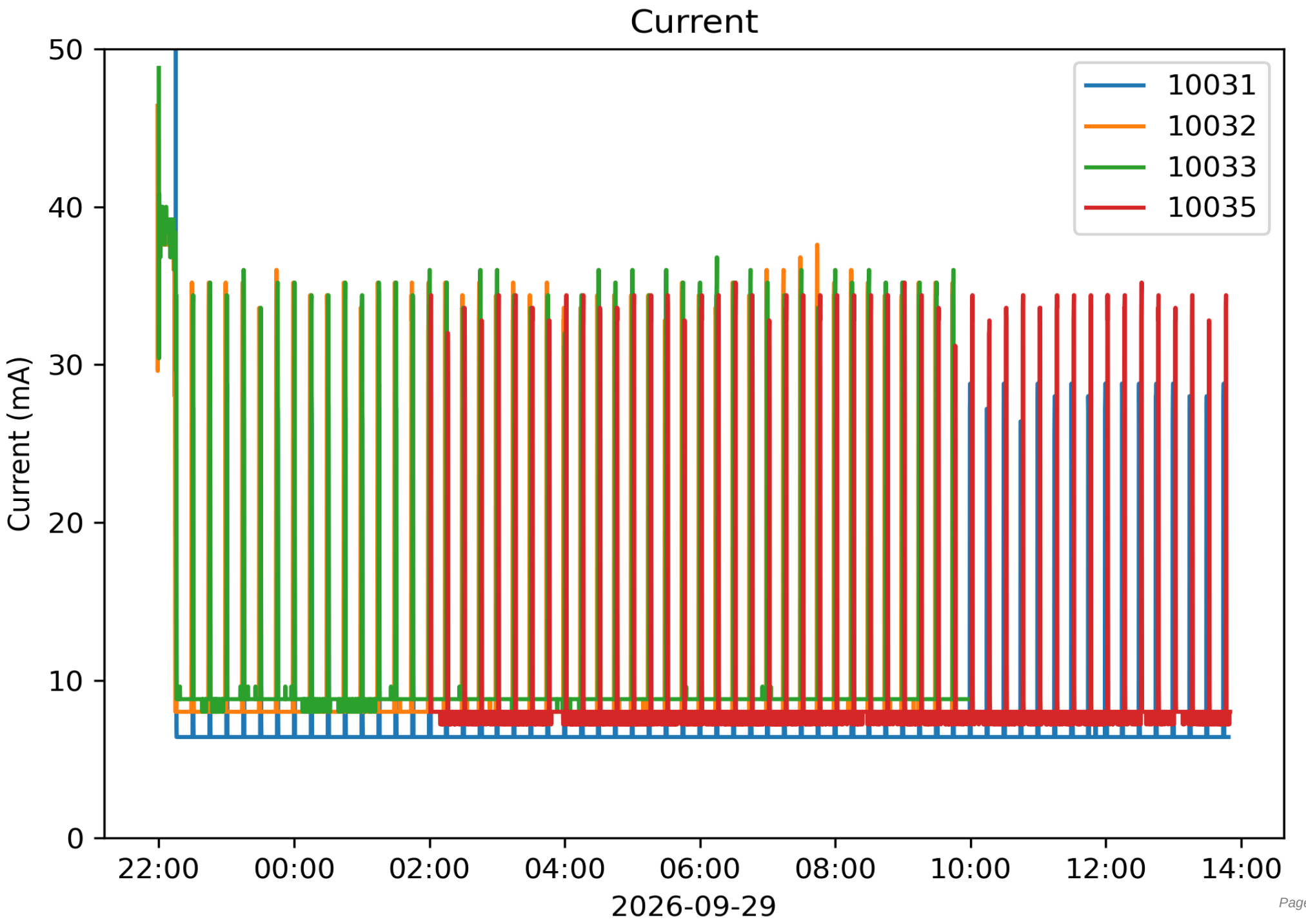
Logger Sensor Status Summary

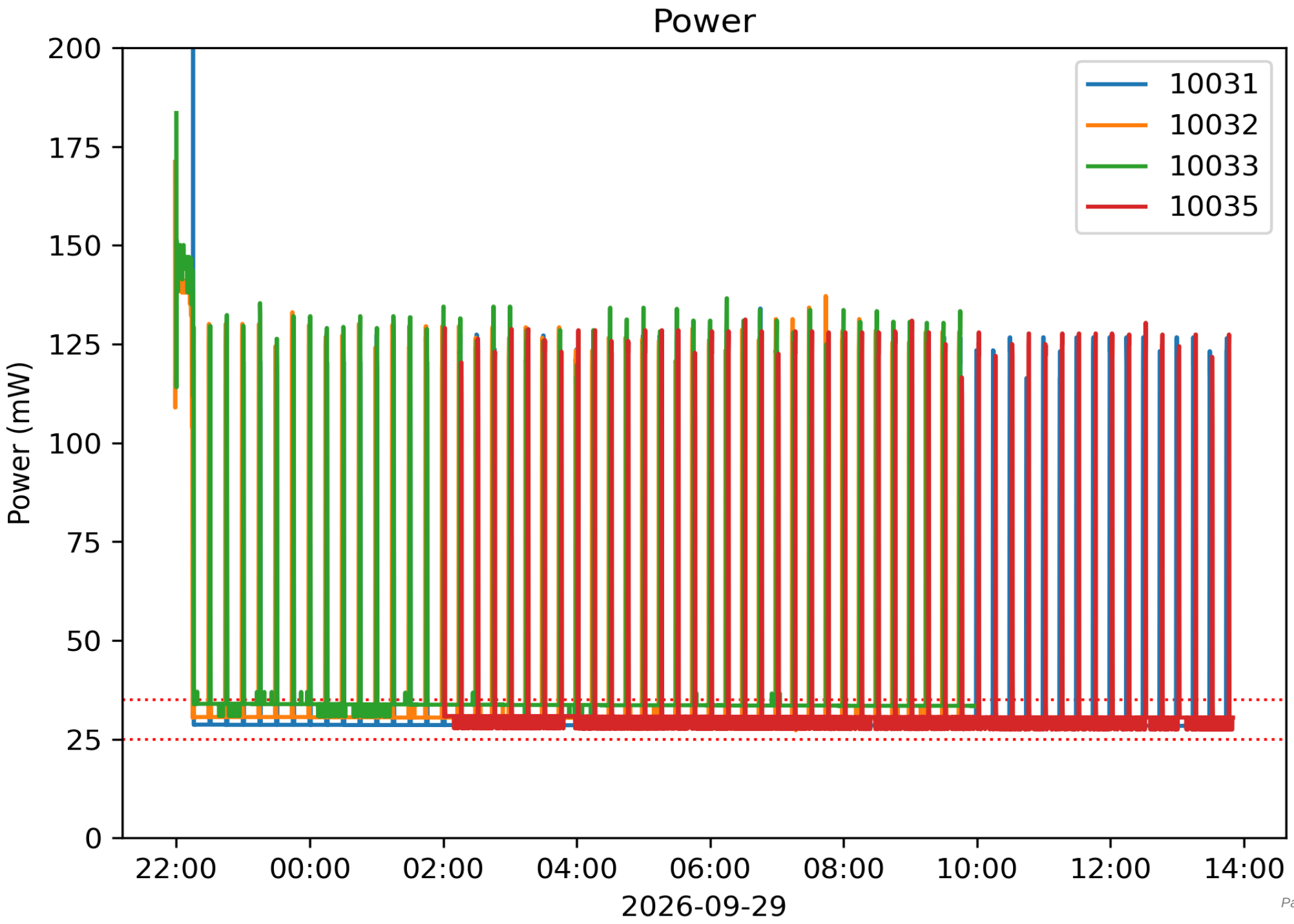
SN	Battery min	Battery max	Temp min	Temp max	mW min	mW max	mA polarity	GPS run time
10031	OKAY	OKAY	ERROR	ERROR	OKAY	OKAY	ERROR	OKAY
10032	OKAY	OKAY	OKAY	OKAY	OKAY	OKAY	ERROR	OKAY
10033	OKAY	OKAY	OKAY	OKAY	OKAY	OKAY	ERROR	OKAY
10035	OKAY	OKAY	OKAY	OKAY	OKAY	OKAY	ERROR	OKAY

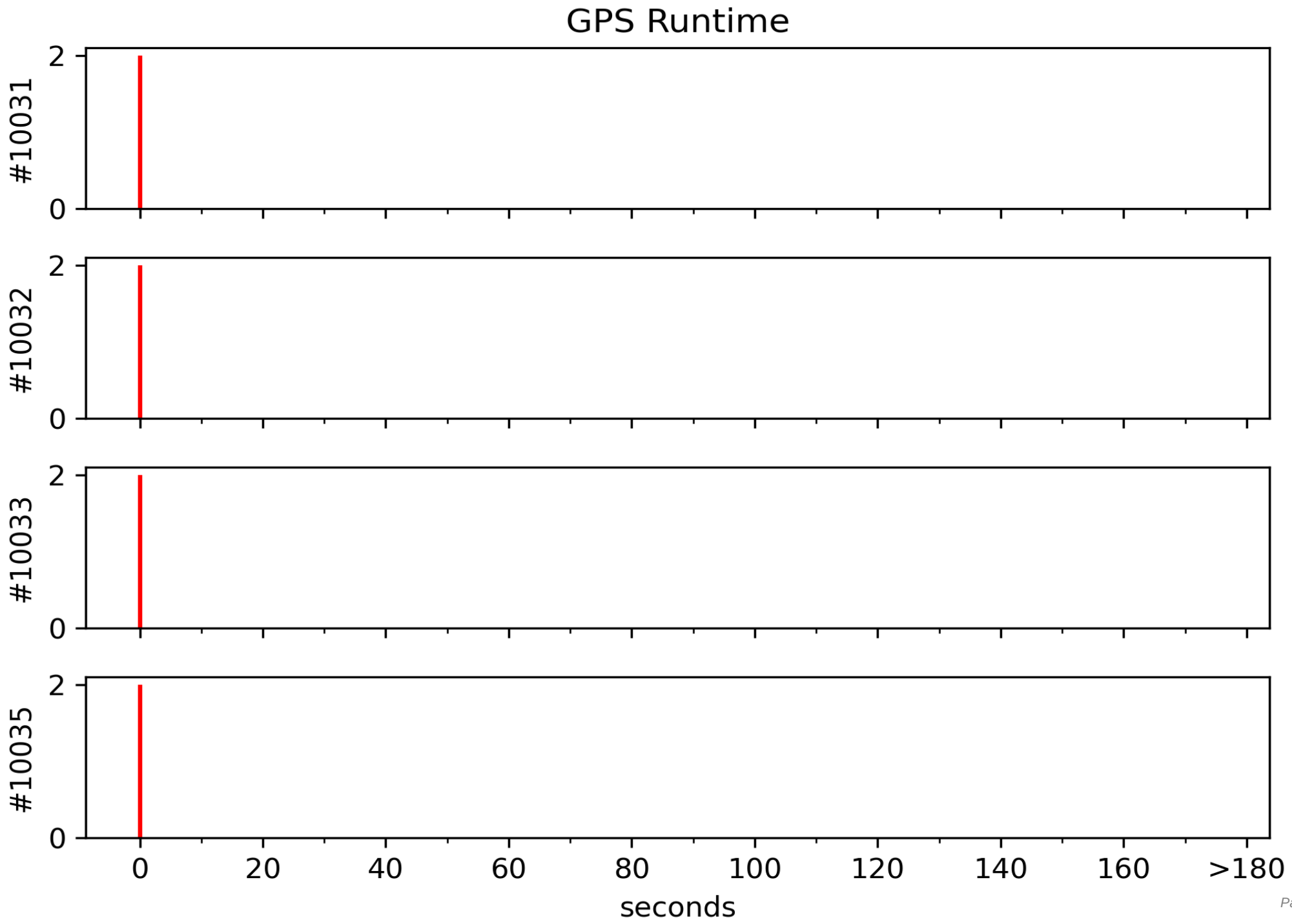
Logger Sensor Metadata Details

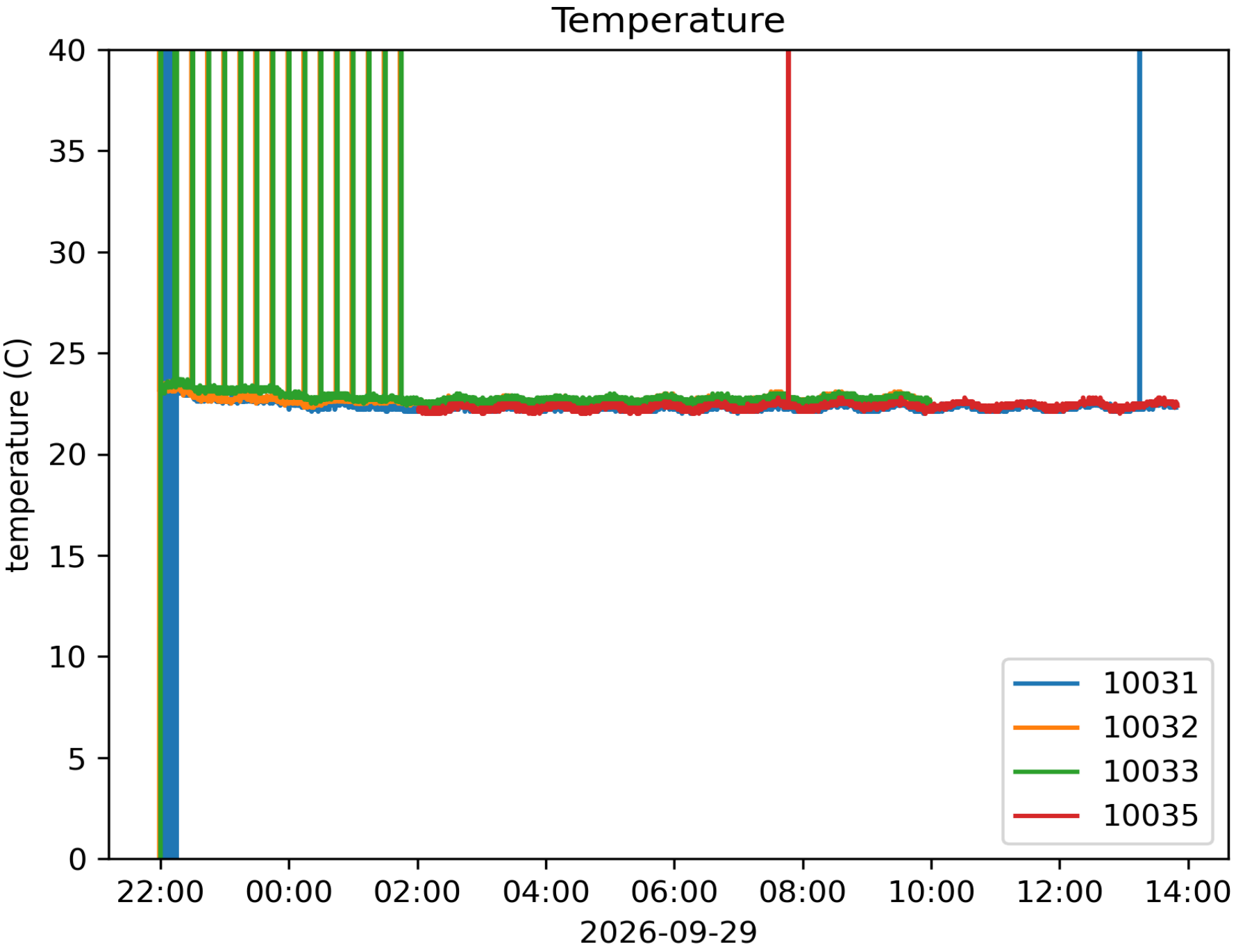
SN	Battery min	Battery max	Temp min	Temp max	Temp average	V min	V max	V average	mW min	mW max	mW average	RH min	RH max	RH average	Mean GPS run time	GPS on-time ratio
10031	4.35	4.51	-44.7	129.6	22.747	0.272	26.896	4.442	24.909	134.003	33.202	3.0	31.0	30.136	nan	0.0
10032	3.57	3.85	22.2	23.6	22.694	3.624	3.832	3.79	27.187	172.563	36.205	3.0	32.0	31.157	nan	0.0
10033	3.66	3.85	22.3	23.8	22.805	3.704	3.856	3.819	30.464	183.488	39.073	3.0	31.0	30.03	nan	0.0
10035	3.66	3.86	21.9	22.9	22.362	3.704	3.856	3.825	27.36	133.92	32.589	3.0	32.0	31.964	nan	0.0

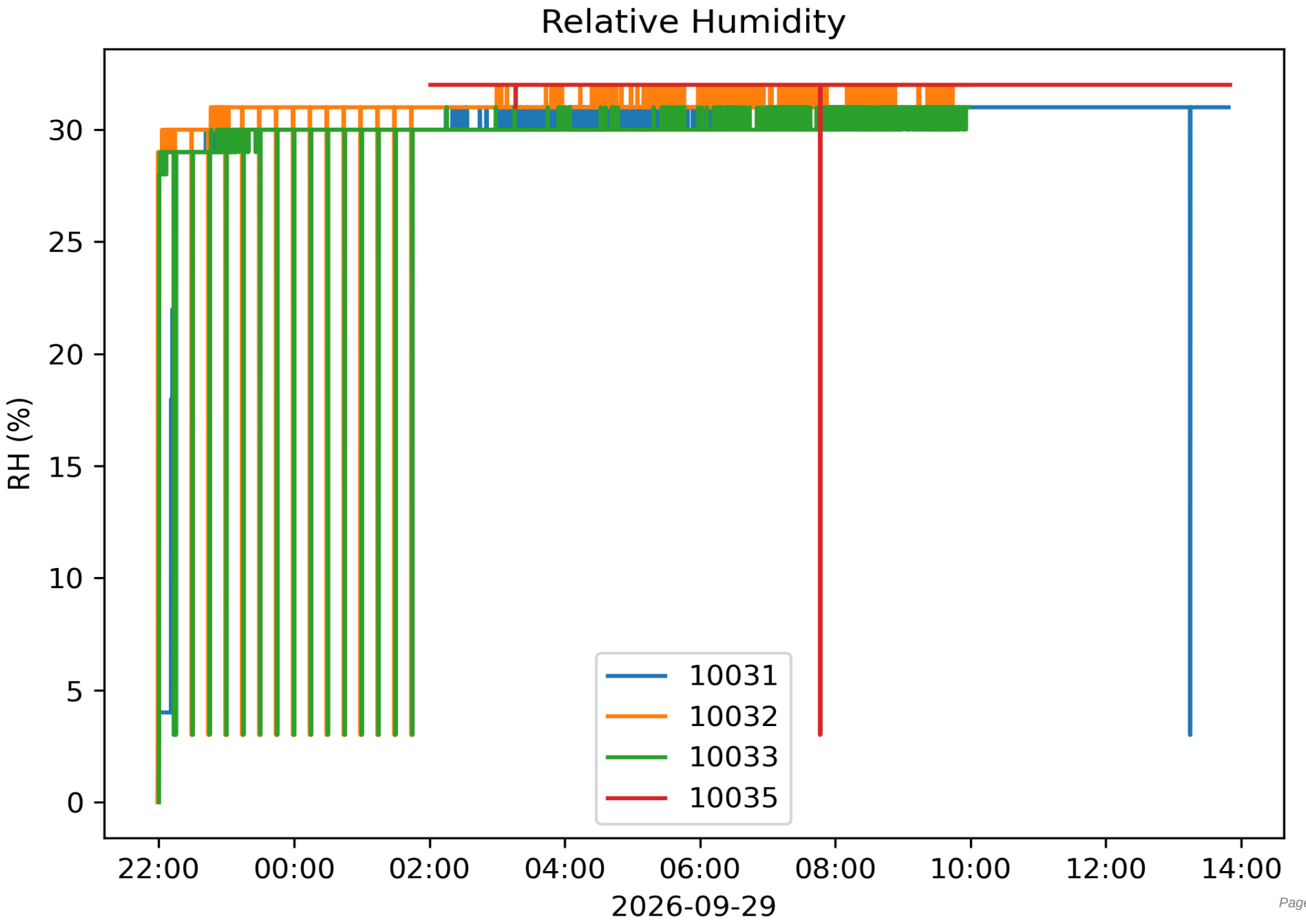

Voltage
16
14
12
10
8
6
4
2
0
voltage (V)
22:00
00:00
02:00
04:00
06:00
08:00
10:00
12:00
14:00
2026-09-29
10031
10032
10033
10035

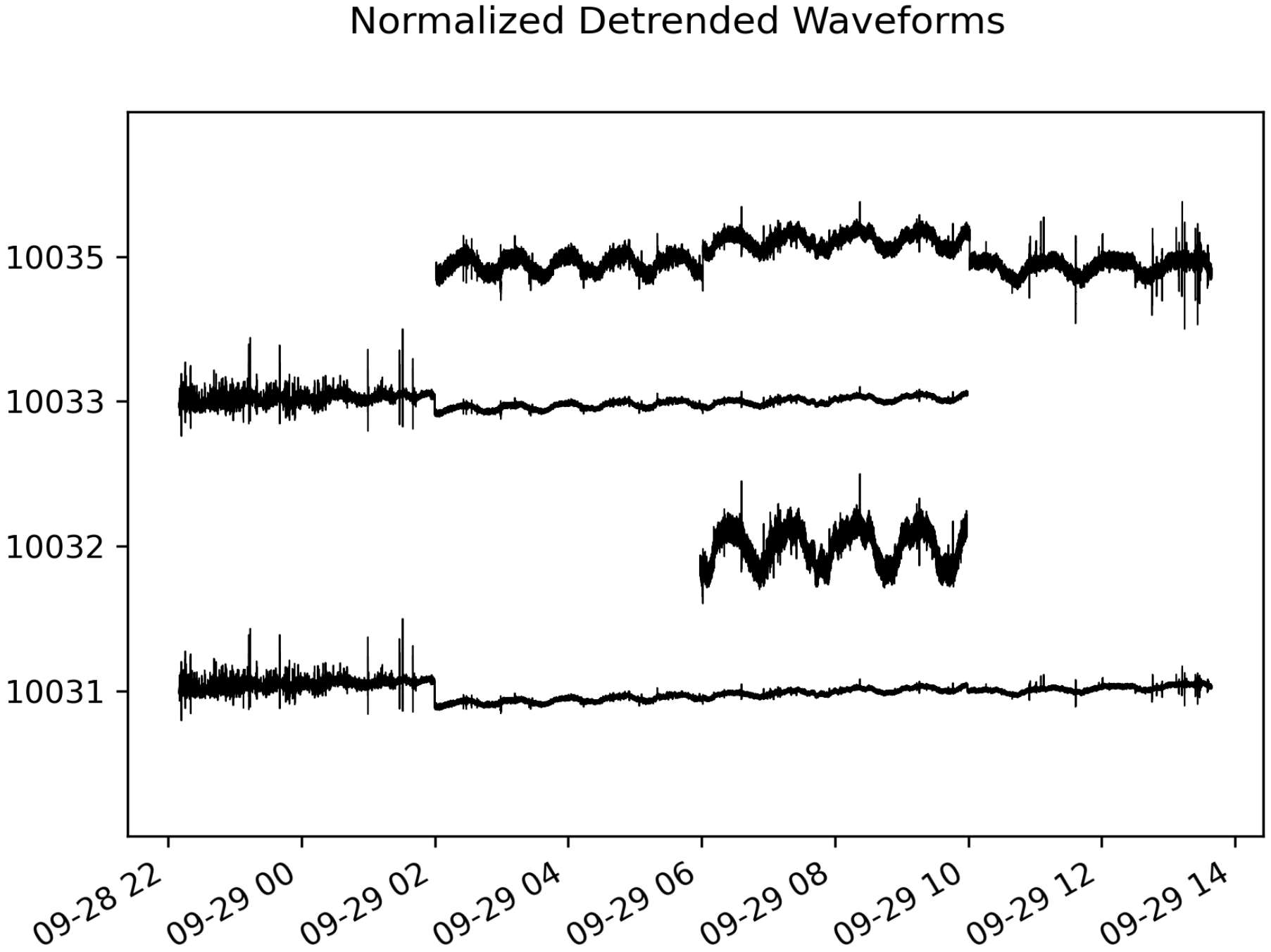

Current
10031
10032
10033
10035
Current (mA)
0
10
20
30
40
50
22:00
00:00
02:00
04:00
06:00
08:00
10:00
12:00
14:00
2026-09-29


Power
Power (mW)
200
175
150
125
100
75
50
25
0
22:00
00:00
02:00
04:00
06:00
08:00
10:00
12:00
14:00
2026-09-29
10031
10032
10033
10035


GPS Runtime
#10031
#10032
#10033
#10035
2
0
0
20
40
60
80
100
120
140
160
>180
seconds


Temperature
temperature (C)
40
35
30
25
20
15
10
5
0
22:00
00:00
02:00
04:00
06:00
08:00
10:00
12:00
14:00
2026-09-29
10031
10032
10033
10035


Relative Humidity
RH (%)
0
5
10
15
20
25
30
22:00
00:00
02:00
04:00
06:00
08:00
10:00
12:00
14:00
2026-09-29
10031
10032
10033
10035


Normalized Detrended Waveforms
10035
10033
10032
10031
09-28 22
09-29 00
09-29 02
09-29 04
09-29 06
09-29 08
09-29 10
09-29 12
09-29 14

Group Test Results

10032 GROUP GPS ERROR: The stop time is 211.7 minutes before the acceptable end time range.
10033 GROUP GPS ERROR: The stop time is 210.9 minutes before the acceptable end time range.
10035 GROUP GPS ERROR: The start time is 222.1 minutes after the acceptable start time range.
SN ['10035'] recorded temperatures greater than 1.5 on either side of the temperature median -45.0 on 2026-09-28 21:58:00. Total temperature range = 67.2 (alpha)
SN ['10031'] recorded temperatures greater than 1.5 on either side of the temperature median 23.1 on 2026-09-28 21:59:00. Total temperature range = 13.8 (alpha)
SN ['10031'] recorded temperatures greater than 1.5 on either side of the temperature median 23.2 on 2026-09-28 22:00:00. Total temperature range = 103.9 (alpha)
SN ['10031'] recorded temperatures greater than 1.5 on either side of the temperature median 23.3 on 2026-09-28 22:01:00. Total temperature range = 40.5 (alpha)
SN ['10031'] recorded temperatures greater than 1.5 on either side of the temperature median 22.7 on 2026-09-28 22:02:00. Total temperature range = 57.9 (alpha)
SN ['10031'] recorded temperatures greater than 1.5 on either side of the temperature median 23.4 on 2026-09-28 22:03:00. Total temperature range = 51.9 (alpha)
SN ['10031'] recorded temperatures greater than 1.5 on either side of the temperature median 23.2 on 2026-09-28 22:04:00. Total temperature range = 100.0 (alpha)
SN ['10031'] recorded temperatures greater than 1.5 on either side of the temperature median 23.3 on 2026-09-28 22:05:00. Total temperature range = 100.7 (alpha)
SN ['10031'] recorded temperatures greater than 1.5 on either side of the temperature median 22.8 on 2026-09-28 22:06:00. Total temperature range = 36.7 (alpha)
SN ['10031'] recorded temperatures greater than 1.5 on either side of the temperature median 23.4 on 2026-09-28 22:07:00. Total temperature range = 93.1 (alpha)
SN ['10031'] recorded temperatures greater than 1.5 on either side of the temperature median 23.4 on 2026-09-28 22:08:00. Total temperature range = 15.2 (alpha)
SN ['10031'] recorded temperatures greater than 1.5 on either side of the temperature median 23.3 on 2026-09-28 22:10:00. Total temperature range = 107.5 (alpha)
SN ['10031'] recorded temperatures greater than 1.5 on either side of the temperature median 23.4 on 2026-09-28 22:11:00. Total temperature range = 102.8 (alpha)
SN ['10031'] recorded temperatures greater than 1.5 on either side of the temperature median 22.6 on 2026-09-28 22:12:00. Total temperature range = 43.5 (alpha)
SN ['10031'] recorded temperatures greater than 1.5 on either side of the temperature median 22.7 on 2026-09-28 22:13:00. Total temperature range = 67.1 (alpha)

Test Info

BATTERY TEST INFO: This test ensures the voltage of each logger is within 1.7 to 15 Volts at each recorded instance. loggers within this specified range, but within a range 0.5 Volts from the threshold will result in a warning message. See battery voltage figure for details.

TEMPERATURE TEST INFO: This test ensures the logger is recording ambient air temperatures within an reasonable range. This range is set at -20 to 60 Celsius or -4.0 to 140.0 Fahrenheit. Temperatures outside this range are considered unlikely for a lab test and could cause malfunctioning or damage.

A2 AND A3 TEST INFO (GEM-ONLY): These tests ensures that the A2 and A3 "auxiliary" inputs on the circuit board are functioning properly. The first test ensures that the metadata has not flatlined for more than 99% of the recorded time (which would indicate a short circuit). The second test ensures it is within a range of 0 - 3.1 V.

FIFO TEST INFO (GEM-ONLY): The FIFO (first-in first-out) is a 75-sample buffer for storing samples while waiting for them to write to the memory card. Use of the FIFO should generally be low. FIFO use should not exceed 5 more than 1% of the time, and should never exceed 50.

OVERRUNS INFO (GEM-ONLY): An overrun is when the FIFO buffer fills and samples are lost. This is uncommon, and recurring overrun issues are particularly uncommon. Any overruns are cause for concern (not alarm) and justify investigating and re-testing a Gem to see whether they persist.

'GPS RUNTIME' and 'GPS ON-TIME RATIO' INFO: These tests ensure that the GPS runs for a reasonable length of time over the course of the test. The first GPS cycle is long (9-15 minutes) due to the need to listen to

a complete almanac transmission from the satellites. Subsequent cycles are just long enough to record 20 GPS fixes; given the 10-30 seconds typically needed to find a fix, the GPS should run for about 30-50 seconds per 15-minute cycle. The 'GPS run time' stat is the average duration of GPS cycles after the first one, and the similar 'GPS ON-TIME RATIO' stat averages the run time of the GPS over the entire recording period in units of seconds per 15-minute GPS cycle.

MA POLARITY TEST INFO (ASPEN-ONLY): This test ensures that most of the current-sensor mA readings are strictly positive. A majority-negative sign suggests the current sensor is wired with reversed polarity.

POWER (MW) TEST INFO (ASPEN-ONLY): This test checks that power (mA * V, excluding spikes above 300 mW) stays within expected bounds: the minimum power should be 20-35 mW, and the maximum power should be 125-200 mW.

RELATIVE HUMIDITY TEST INFO (ASPEN-ONLY): This reports the minimum, maximum, and average relative humidity recorded by the Aspen board's humidity sensor, excluding 0% readings (a known sensor-glitch sentinel value). There is currently no pass/fail threshold; it is informational.

Test Troubleshooting

BATTERY TROUBLESHOOTING: If you received a battery error or warning, either the battery voltage is outside the specified range and the logger managed to record anyway (possibly because the battery died during the test), or the battery voltage sensor is malfunctioning. If the actual battery voltage does not match what was recorded, there may be a bad connection involving the battery voltage sensor (R8 and R9) or pin A1.

TEMPERATURE TROUBLESHOOTING: If you received a temperature warning, you are approaching within 5 C the limit of the temperature range operation for the logger (-20 to 60 C). If this value does not reflect an accurate ambient air temperature, the TMP36 sensor on the board (U4) may be damaged, or there may be a bad connection involving the sensor or pin A0.

A2 AND A3 TROUBLESHOOTING: If you consistently see A2 or A3 flatlined, it may mean that those pins have a short circuit.

FIFO TROUBLESHOOTING: FIFO-related problems are unusual and generally do not disqualify a Gem from service in the absence of observed malfunctioning. Consistent FIFO problems could be due to communication problems with the memory card. In this case, try a different card (SanDisk brand strongly preferred). If that doesn't fix it, the problem may be due to poor connections between the microcontroller and card, or problems with the microcontroller. This is all speculative because the problem is not known to have occurred and has not been studied.

OVERRUNS TROUBLESHOOTING: Overrun problems are a more serious symptom of the same problems that cause FIFO issues. Consistent overrun problems could be due to communication problems with the memory card. In this case, try a different card (SanDisk brand strongly preferred). If that doesn't fix it, the problem may be due to poor connections between the microcontroller and card, or problems with the microcontroller. This is all speculative because overruns are uncommon and have not been studied in any detail.

GPS TROUBLESHOOTING: Ensure that the test site has good GPS signal (many sites that are secure and convenient for testing have very poor views of the sky). A GPS with a fix blinks red briefly once every 15 seconds, and the logger connected to it should be blinking blue 3x/second if it's getting all the information from the GPS. If GPS signal is strong but a logger is running for long periods of time, a communication issue between GPS and microcontroller is the most likely explanation. Check that all GPS terminal screws are tight and that the GPS does not somehow have a short circuit to another component. There may be a poor connection along the three GPS communication pins.

MA POLARITY TROUBLESHOOTING: If most mA readings are negative, check the wiring/orientation of the current sensor on the Aspen board.

POWER (MW) TROUBLESHOOTING: A minimum or maximum power outside the expected range may indicate the logger is drawing abnormal current (e.g. due to a hardware fault) or that the voltage/current sensor readings are unreliable.

RELATIVE HUMIDITY TROUBLESHOOTING: Humidity values outside a plausible range for the test environment may indicate a malfunctioning humidity sensor.